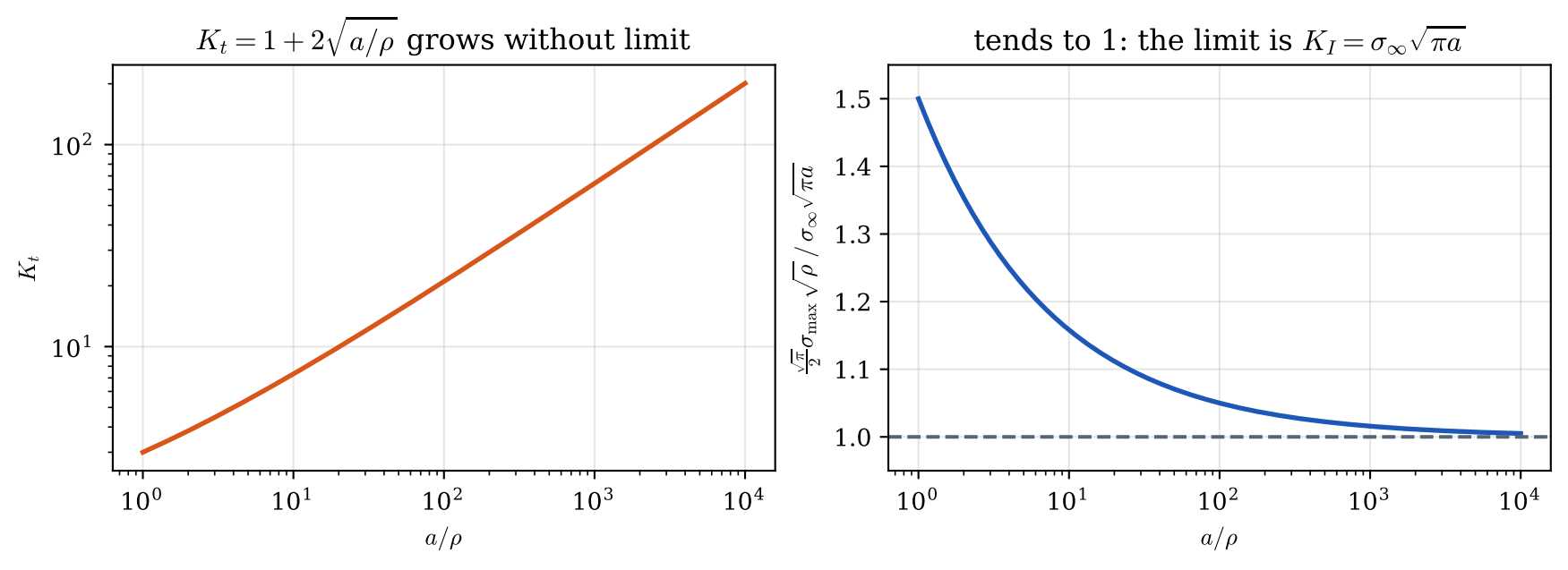
$K_t = 1 + 2\sqrt{a/\rho}$ grows without limit
K_t
10^2
10^1
10^0
10^1
10^2
10^3
10^4
a/ρ
tends to 1: the limit is $K_I = \sigma_\infty \sqrt{\pi a}$
$\frac{\sqrt{\pi}}{2}\sigma_{\max}\sqrt{\rho}\,/\,\sigma_\infty\sqrt{\pi a}$
1.5
1.4
1.3
1.2
1.1
1.0
10^0
10^1
10^2
10^3
10^4
a/ρ